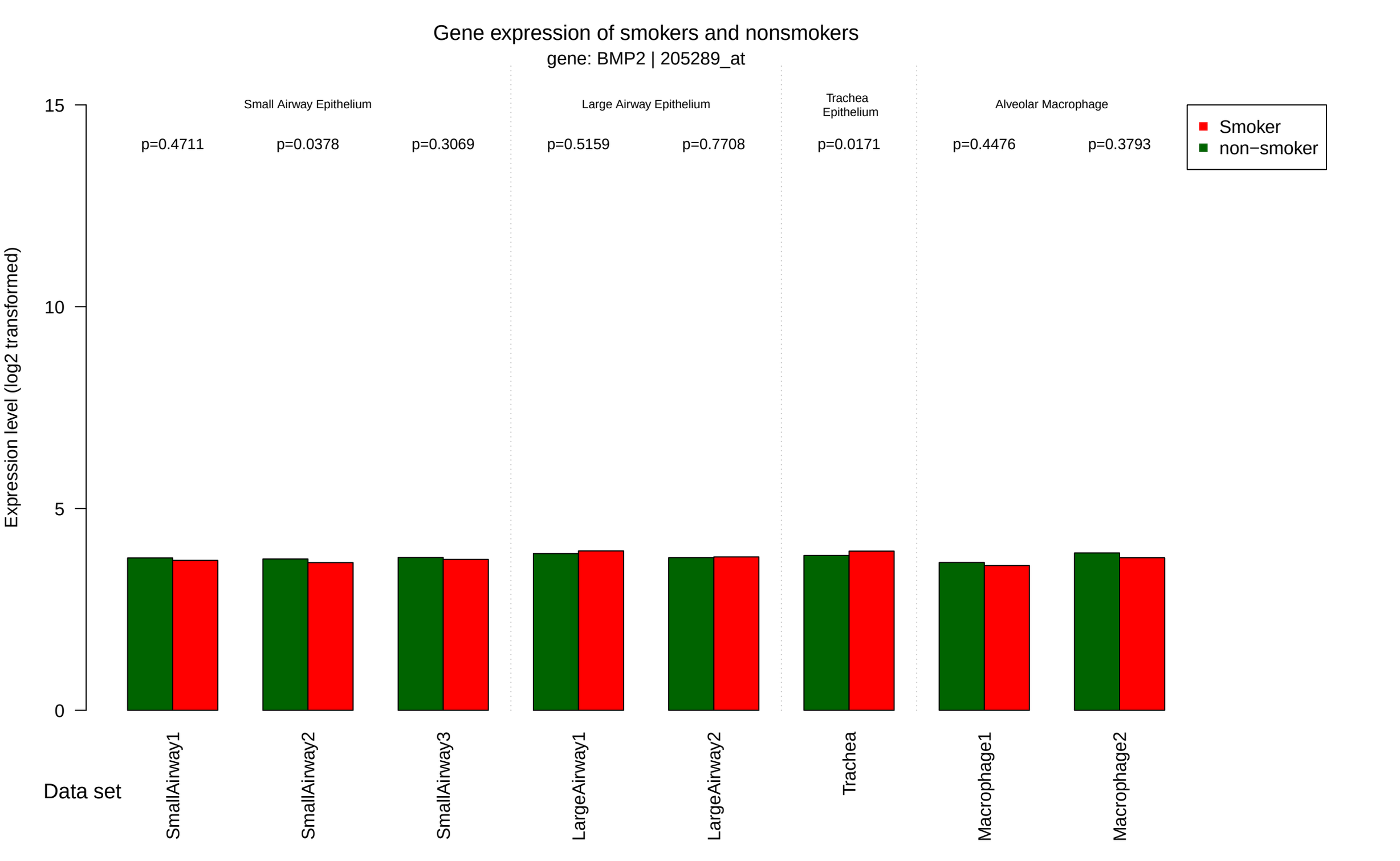
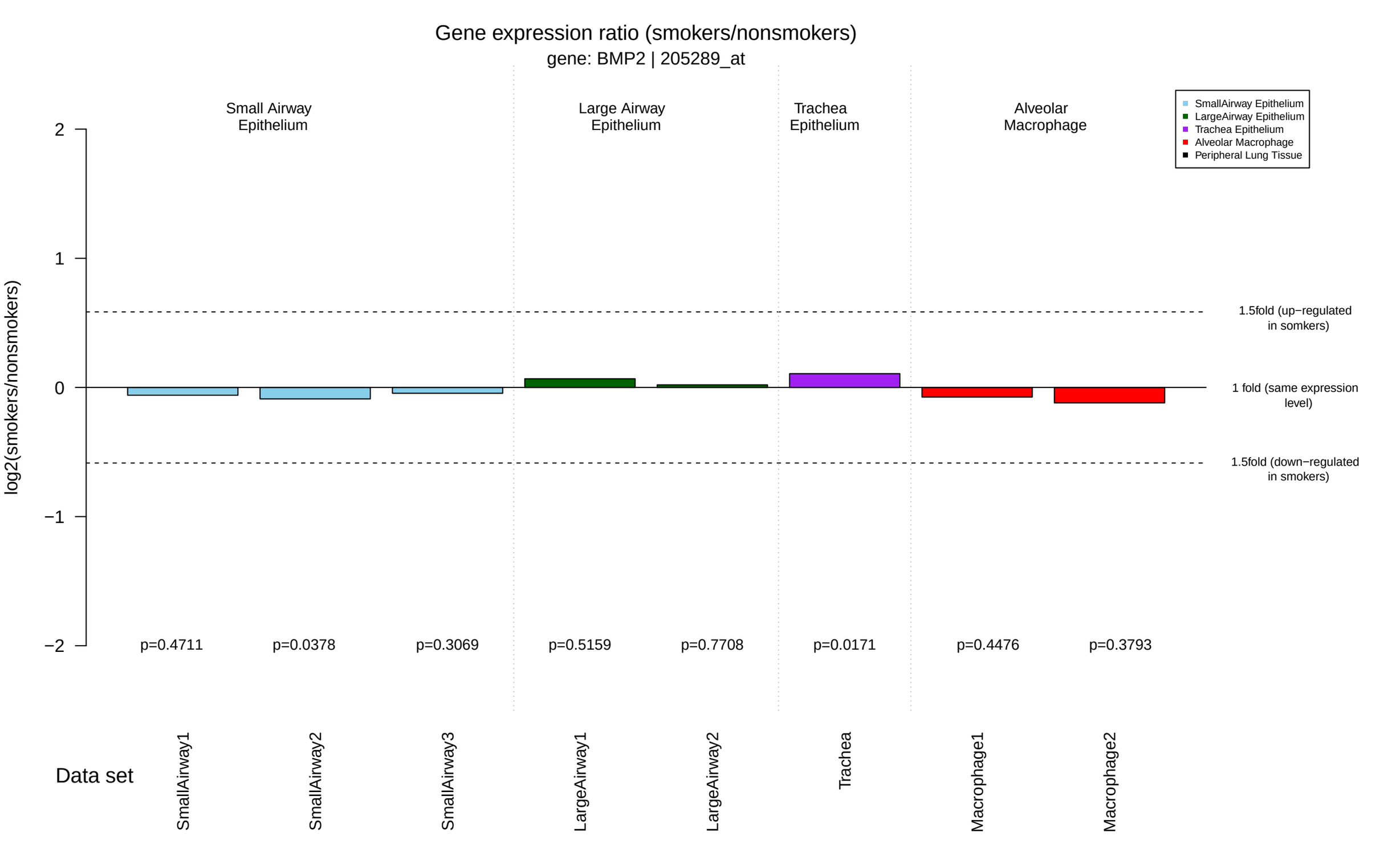
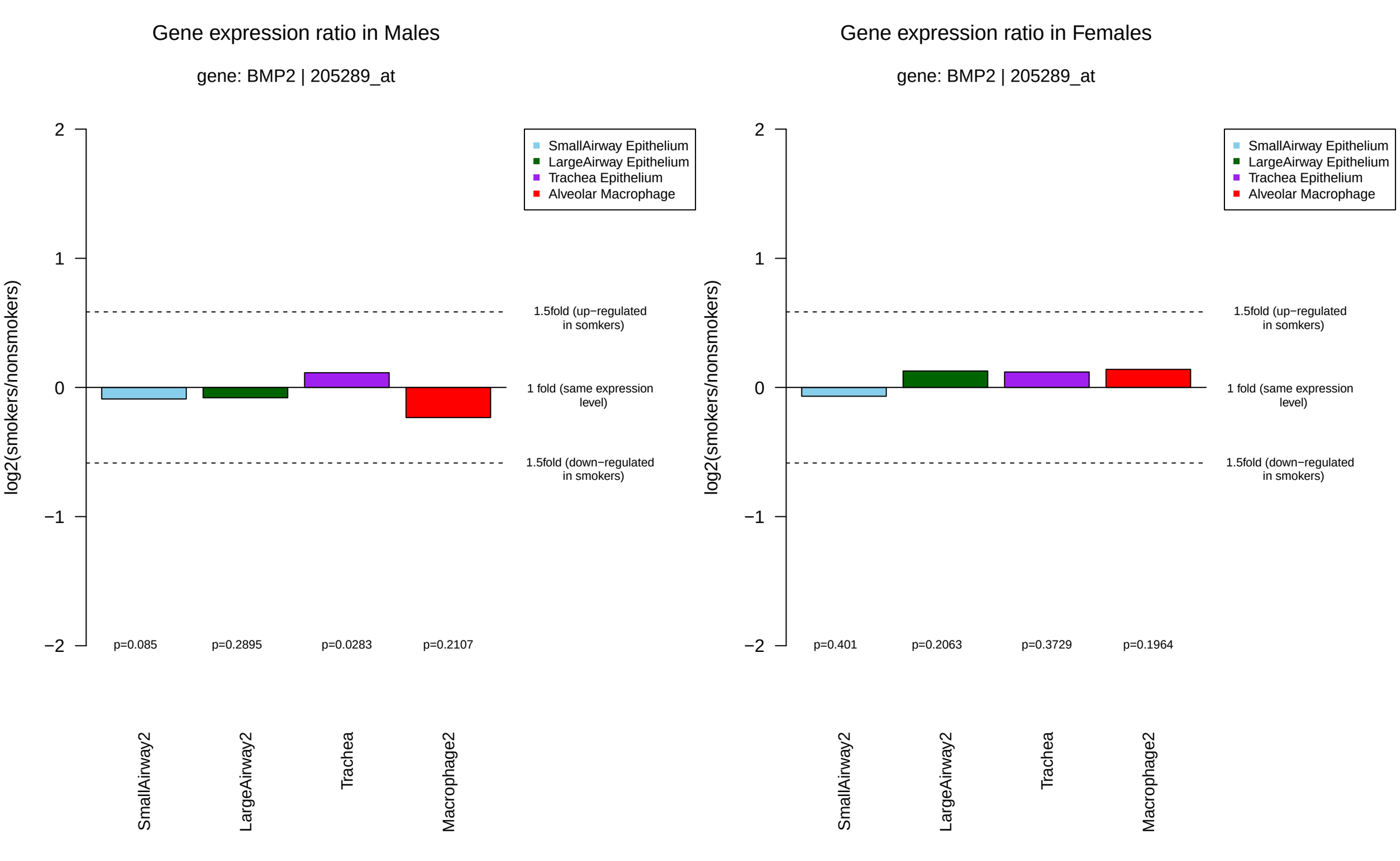
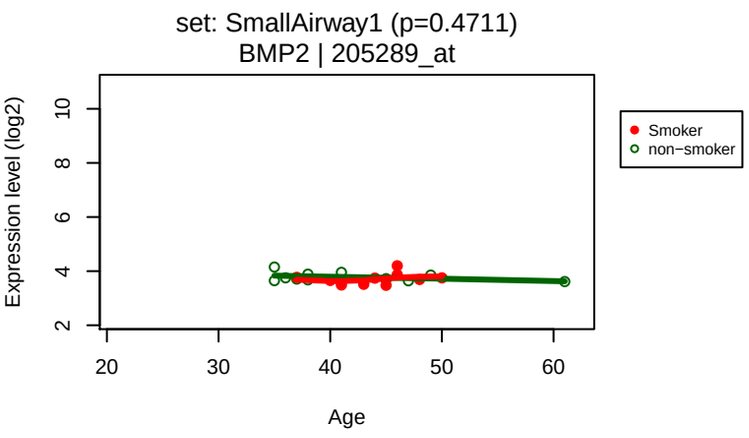
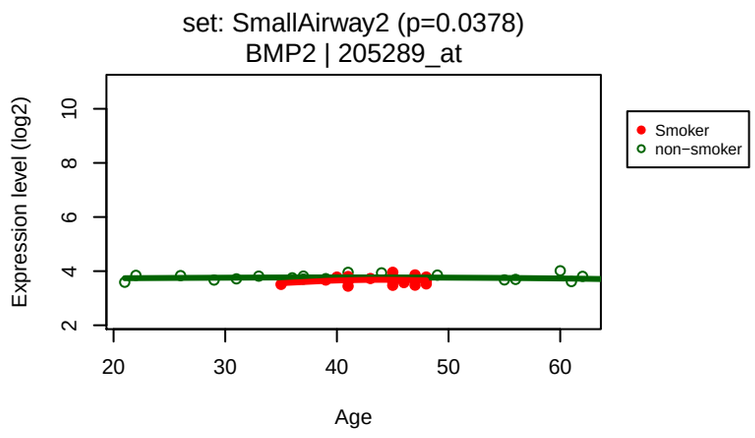
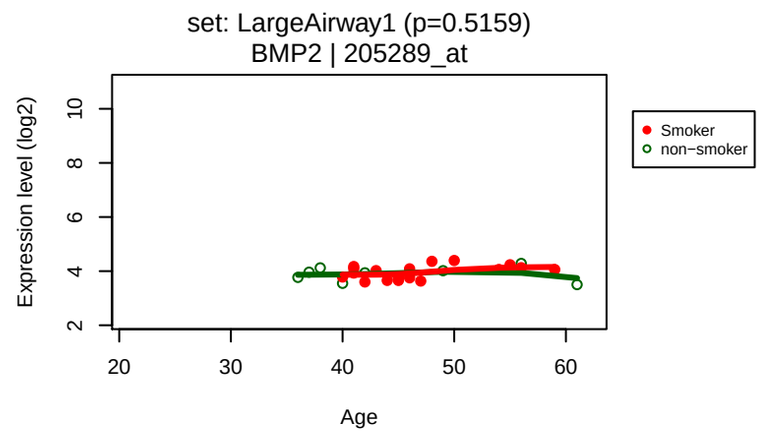

Gene expression of smokers and nonsmokers
gene: BMP2 | 205289_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4711
p=0.0378
p=0.3069
p=0.5159
p=0.7708
p=0.0171
p=0.4476
p=0.3793
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

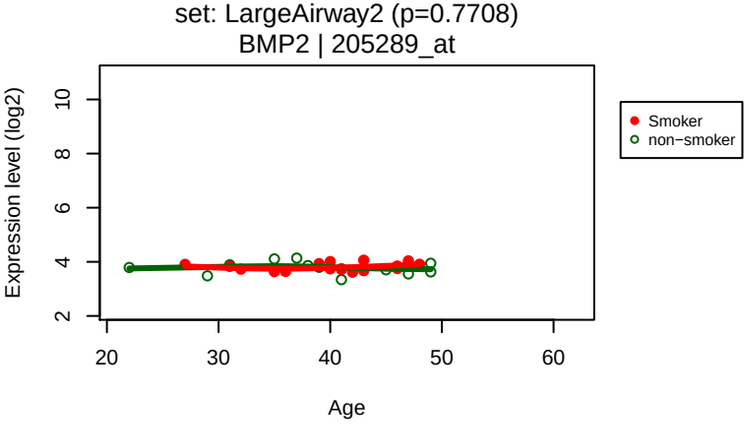

Gene expression ratio (smokers/nonsmokers)
gene: BMP2 | 205289_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4711
p=0.0378
p=0.3069
p=0.5159
p=0.7708
p=0.0171
p=0.4476
p=0.3793
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

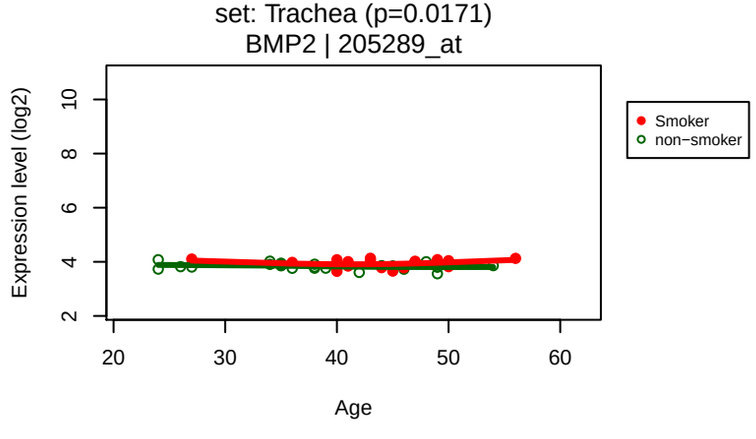

Gene expression ratio in Males
gene: BMP2 | 205289_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.085
p=0.2895
p=0.0283
p=0.2107
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BMP2 | 205289_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.401
p=0.2063
p=0.3729
p=0.1964
SmallAirway2
LargeAirway2
Trachea
Macrophage2

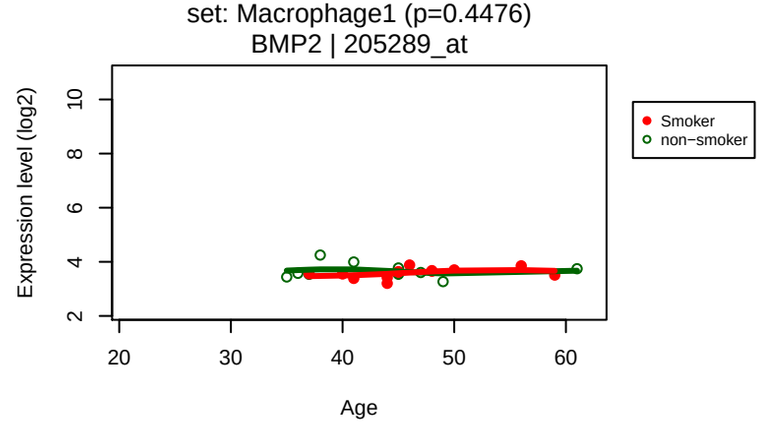

set: SmallAirway1 (p=0.4711)
BMP2 | 205289_at
Expression level (log2)
Age
Smoker
non−smoker

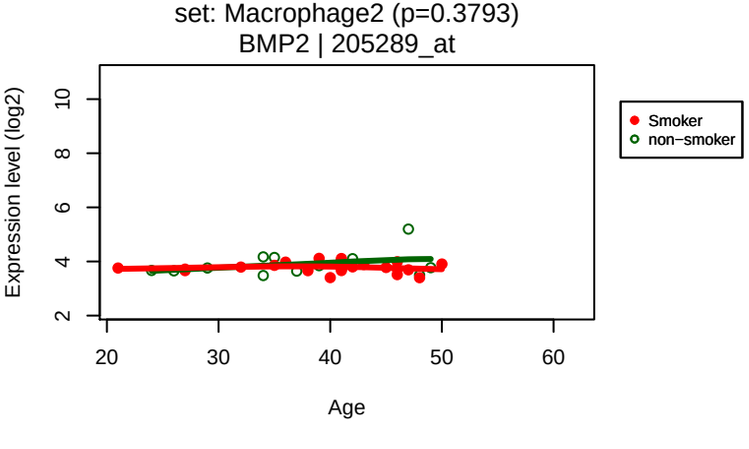

set: SmallAirway2 (p=0.0378)
BMP2 | 205289_at
Expression level (log2)
Age
Smoker
non−smoker

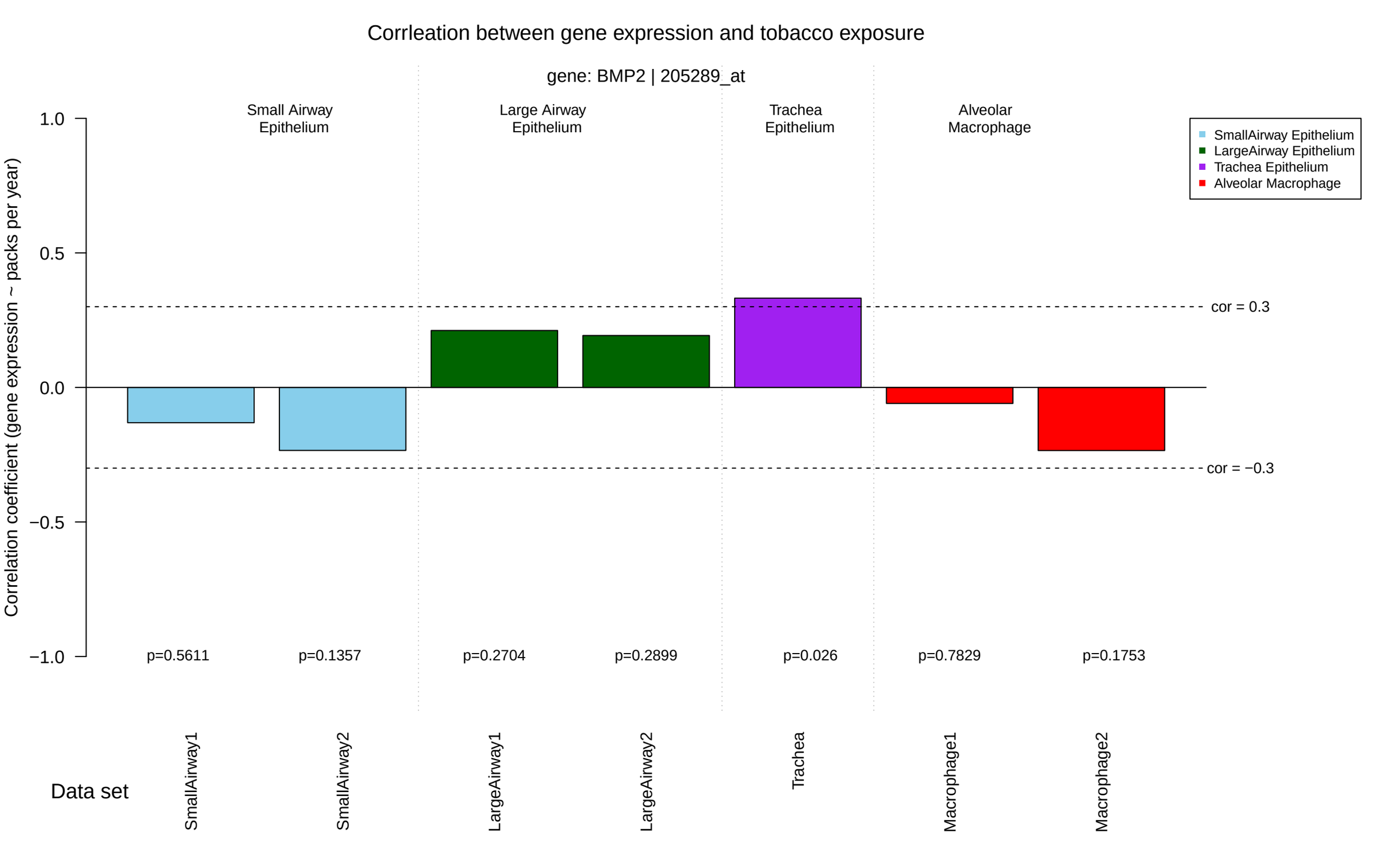

set: LargeAirway1 (p=0.5159)
BMP2 | 205289_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7708)
BMP2 | 205289_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0171)
BMP2 | 205289_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4476)
BMP2 | 205289_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3793)
BMP2 | 205289_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BMP2 | 205289_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5611
p=0.1357
p=0.2704
p=0.2899
p=0.026
p=0.7829
p=0.1753
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2